Literature Status: Discrete Supports for fsJN-Sequences

Automatic Math Discovery Run `fa_banach_001`

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Status

This is a **literature_already_answered** packet. It records that a later paper explicitly answers one of the open questions extracted from arXiv:2009.07552. It is not a new proof packet.

Source Question

The source paper is:

> Jerzy Kąkol, Damian Sobota, and Lyubomyr Zdomskyy, *The Josefson–Nissenzweig theorem, Grothendieck property, and finitely supported measures on compact spaces*, arXiv:2009.07552.

In Question 4.23, on page 24 of the source PDF, the authors ask:

> Does every space X with the fsJNP admit an fsJN-sequence $\langle \mu_n : n \in \omega \rangle$ such that $\bigcup_n \operatorname{supp}(\mu_n)$ is a discrete subspace of X?

Here fsJNP means the finitely supported Josefson–Nissenzweig property, and an fsJN-sequence is a weak-star null normalized sequence of finitely supported measures.

Supporting Answer

The supporting paper is:

> Witold Marciszewski, Damian Sobota, and Lyubomyr Zdomskyy, *On sequences of finitely supported measures related to the Josefson–Nissenzweig theorem*, arXiv:2303.03809.

Theorem 1.2 of the supporting paper, on page 2, states that if a Tychonoff space X carries a finitely supported JN-sequence $\langle \mu_n : n \in \omega \rangle$, then there exists a finitely supported JN-sequence $\langle \nu_n : n \in \omega \rangle$ on X such that:

$$\bigcup_{n\in\omega} \operatorname{supp}(\nu_n) \subseteq \bigcup_{n\in\omega} \operatorname{supp}(\mu_n),$$

$$\operatorname{supp}(\nu_n) \cap \operatorname{supp}(\nu_k) = \varnothing \quad (n \neq k),$$

and

$$\bigcup_{n\in\omega} \operatorname{supp}(\nu_n) \text{ is a discrete subset of } X.$$

The proof is given by combining Theorem 4.3, which produces disjoint supports, with Theorem 5.4, which refines a disjointly supported sequence so that the union of supports is discrete.

Identification

The source question asks for exactly the existence of an fsJN-sequence with discrete union of supports under the hypothesis that X has the fsJNP. Theorem 1.2 of arXiv:2303.03809 proves this for every Tychonoff space X, and hence gives a positive answer to Question 4.23 of arXiv:2009.07552.

The supporting arXiv record also notes substantial text overlap with arXiv:2009.07552, and two of the three supporting authors are coauthors of the source paper, so this is best classified as an already-known later answer rather than as a newly discovered proof.

Scope Limitations

This packet answers only the discrete-support question, Question 4.23, from the source paper. It does not address the source paper's questions about Boolean subalgebras with the ℓ_1-Grothendieck property, long τ-simple extension systems, or de la Vega systems.

Search Evidence

Local cheap indexes for arXiv:2009.07552, fsJNP, finitely supported Josefson–Nissenzweig sequences, discrete supports, and the Grothendieck keywords had no prior exact packet at reservation time. External search on 2026-06-14 for the exact phrase around "fsJNP" and "discrete subspace" returned arXiv:2303.03809 as the direct answer source.

References

[1] J. Kąkol, D. Sobota, and L. Zdomskyy, *The Josefson–Nissenzweig theorem, Grothendieck property, and finitely supported measures on compact spaces*, arXiv:2009.07552.

[2] W. Marciszewski, D. Sobota, and L. Zdomskyy, *On sequences of finitely supported measures related to the Josefson–Nissenzweig theorem*, arXiv:2303.03809.